

MACHINE LEARNING PROJECT REPORT

Title

Customer Retail Data Analysis and Model Comparison using Machine Learning.

Abstract

This project focuses on implementing and comparing multiple Machine Learning algorithms using a customer retail dataset. The dataset contains information such as Quantity, Unit Price, and Country. Different classification models are trained to predict customer country categories based on purchasing behavior. The project includes data preprocessing, visualization, model training, evaluation using confusion matrices and accuracy scores, and graphical comparison of algorithms.

Introduction

Machine Learning is a branch of Artificial Intelligence that enables systems to learn patterns from data and make predictions automatically. In retail industries, Machine Learning helps businesses analyze customer behavior, purchasing patterns, and sales trends.

In this project, three supervised Machine Learning algorithms are implemented:

- Logistic Regression
- Decision Tree Classifier
- K-Nearest Neighbors (KNN)

The performance of these models is evaluated and compared using a retail customer dataset.

Objectives

- To understand the Machine Learning workflow
 - To preprocess and clean retail customer data
 - To visualize relationships between dataset features
 - To implement multiple classification algorithms
 - To compare model performances using accuracy and confusion matrix
 - To understand prediction and evaluation techniques
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Dataset Description

The dataset used is a customer retail dataset containing sales information.

Features Used

Feature	Description
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Quantity	Number of products purchased
UnitPrice	Price per product
Country	Customer country

Technologies Used

- Python
- Pandas
- Matplotlib
- Scikit-learn

Machine Learning Models Used

1. Logistic Regression

A statistical classification algorithm used for predicting categorical outcomes.

Advantages:

- Simple and fast
- Good for linear data

Disadvantages:

- Poor performance on complex datasets

2. Decision Tree Classifier

A tree-based classification algorithm that divides data into branches.

Advantages

- Easy to understand
- Handles nonlinear data

Disadvantages

- Can overfit the data

3. K-Nearest Neighbours (KNN)

A distance-based classification algorithm that classifies data based on nearest neighbours.

Advantages

- Simple implementation
- Good accuracy for small datasets

Disadvantages

- Slow for large datasets
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Methodology

Step 1: Load Dataset

The dataset is loaded using Pandas library.

```
df = pd.read_csv("customer_retail csv file.csv")
```

Step 2: Handle Missing Values

Missing values are removed using:

```
df = df.dropna()
```

Step 3: Encode Categorical Data

Country column is converted into numerical form using LabelEncoder.

```
encoder = LabelEncoder()
```

```
df['Country_encoded'] = encoder.fit_transform(df['Country'])
```

Step 4: Data Visualization

Scatter plot is used to visualize Quantity vs UnitPrice.

```
plt.scatter(df['Quantity'], df['UnitPrice'])
```

Data Visualization

Scatter Plot

The scatter plot shows the relationship between:

- Quantity purchased
- Product Unit Price

Observation

Different purchasing patterns can be identified based on pricing and quantity distribution.

Train-Test Splitting

Dataset is divided into:

- 80% Training Data
- 20% Testing Data

```
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42)
```

Model Training and Testing

Logistic Regression

```
log_model = LogisticRegression()
log_model.fit(x_train, y_train)
```

Decision Tree

```
dt_model = DecisionTreeClassifier()
dt_model.fit(x_train, y_train)
```

KNN

```
knn_model = KNeighborsClassifier(n_neighbors=5)
knn_model.fit(x_train, y_train)
```

Evaluation Metrics

1. Accuracy Score

Measures how many predictions are correct.

Formula

Accuracy = Correct Predictions / Total Predictions

2. Confusion Matrix

Used to evaluate classification performance.

It shows:

- True Positive
 - False Positive
 - True Negative
 - False Negative
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Model Comparison

The performances of Logistic Regression, Decision Tree, and KNN are compared using:

- Accuracy scores
- Confusion matrices
- Bar graph visualization

```
plt.bar(models, accuracies)
```

Results

- Logistic Regression provides baseline classification performance.
- Decision Tree performs well on nonlinear data.
- KNN gives competitive accuracy depending on nearest neighbors.

The comparison graph helps identify the best-performing algorithm.

Learning Outcomes

- Understanding Machine Learning workflow
 - Data preprocessing techniques
 - Feature encoding methods
 - Data visualization
 - Model training and testing
 - Performance evaluation using confusion matrix and accuracy
 - Comparison of classification algorithms
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Advantages of the Project

- Easy implementation
 - Demonstrates supervised learning concepts
 - Useful for retail customer analysis
 - Helps understand model comparison
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Limitations

- Limited dataset features
 - Accuracy depends on dataset quality
 - KNN becomes slower for large datasets
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Future Scope

- Use larger datasets
- Apply Deep Learning models
- Improve feature engineering

- Deploy model using Flask or Streamlit
 - Add advanced visualization dashboards
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Conclusion

This project successfully demonstrates the implementation and comparison of multiple Machine Learning classification algorithms using a customer retail dataset. The project highlights the importance of preprocessing, visualization, training, evaluation, and model comparison in Machine Learning applications. It also provides practical understanding of supervised learning techniques and performance analysis.